Name- Ayush Singh

Course Name – Data Analytics With GenAI

Subject – PowerBI

Topic – Advanced Visuals in PowerBI

Contact Number – 7054891639

Email ID – as764994@gmail.com

PowerBI File : File Link

Q1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

Answer 1.
In Power BI, we have different ways to filter data depending on how we want to use it.
Basic Filter is the simplest one. It's available in the Filters pane and works by letting you check/uncheck specific values. For example, if I want to show only 2023 data, I just tick "2023" from the list. It's straightforward but not very flexible.

Advanced Filter is also in the Filters pane but gives more control. Instead of just selecting values, you can write conditions like "Sales is greater than 50000" or "Product name contains Lap". You can also combine two conditions using AND/OR. It's useful when you need logic-based filtering.

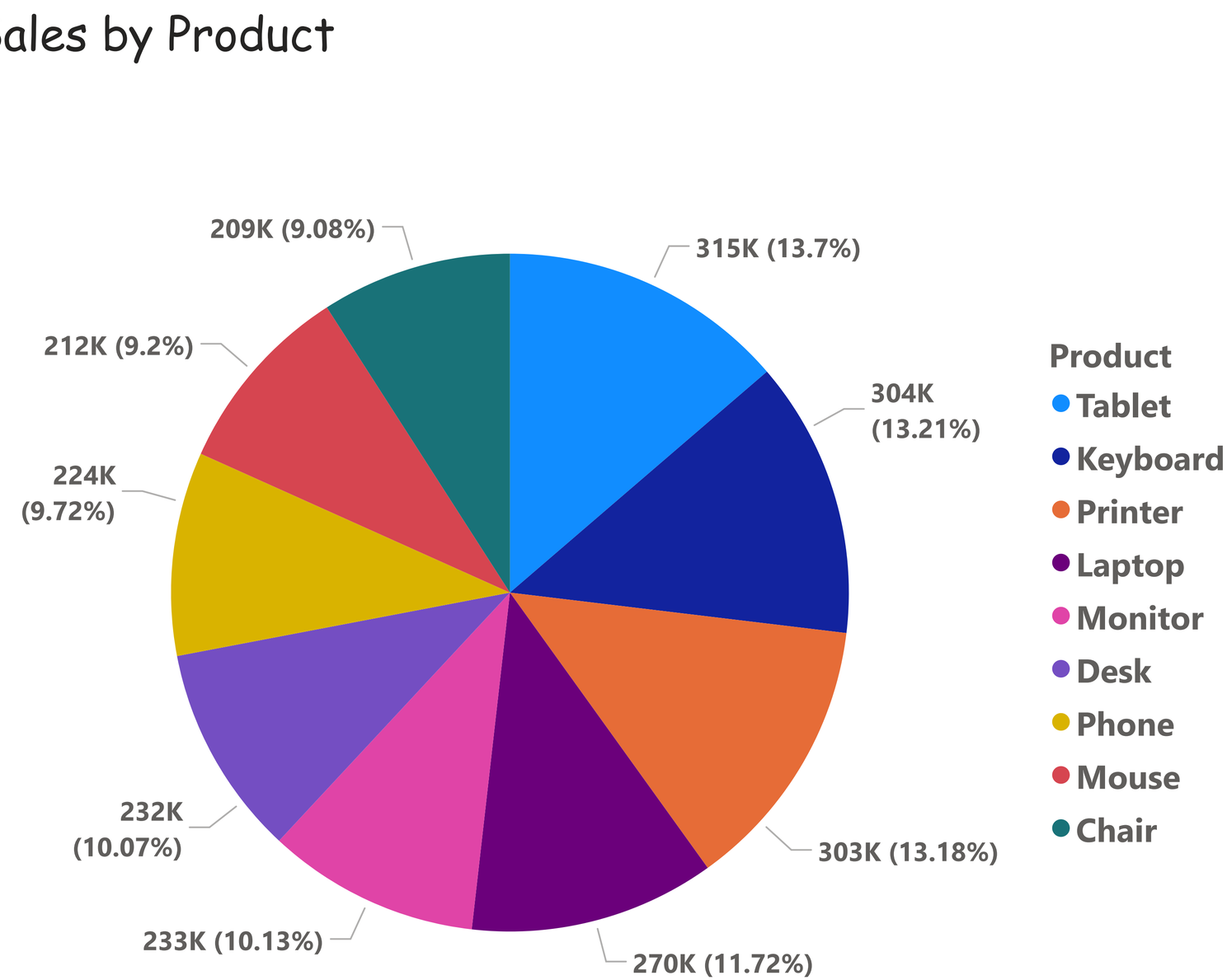
Slicer is different from both, it's an actual visual that sits on the report canvas. Users can see it and interact with it directly. For example, a dropdown or list showing all regions, and clicking "North" filters the whole page. It's the most user-friendly option because non-technical users can also use it without going into the Filters pane.

So basically, Basic Filter is quick and simple, Advanced Filter is for conditional logic, and Slicer is for end-user interactivity on the dashboard itself.

Q2. Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023.

Region	Sum of Sales
East	71681
North	118922
South	264205
West	272091
Total	**726899**

Q3. Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by Region.



Q4. What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

Answer 4

So basically in Power BI, filters work at three different levels depending on how widely you want them to apply.

Visual-Level Filter only affects the one visual you apply it to. Like if I have 3 charts on a page and I apply a filter on one bar chart to show only Electronics, the other two charts are completely unaffected. It's useful when you want one specific visual to show different data than the rest.

Page-Level Filter applies to everything on that particular page. So if I set Country = India on page 1, all visuals on page 1 will show only India data, but page 2 and page 3 are not affected at all.

Report-Level Filter is the broadest one. Whatever filter you set here applies to every single visual on every single page of the report. For example in our Sales dataset, if I set Country = India as a report-level filter, even if I go to page 3, it will still show only India records.

A simple way I remember it — Visual is the smallest scope, Page is medium, and Report is the widest. You choose based on how broadly you want the filter to work.

Q5. Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India.

Country	Sum of Sales
India	2300843

Q6. What is the role of Drill-Down in Power BI? How does it improve data exploration?

Answer 6

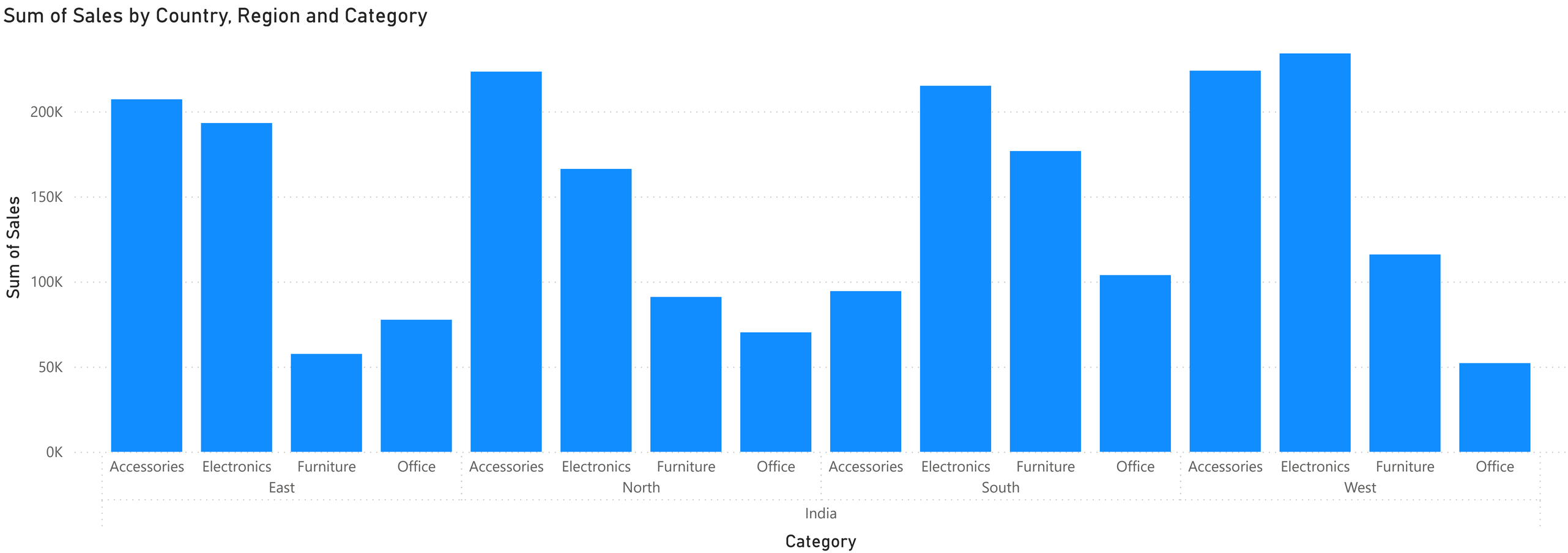
Drill-Down allows users to navigate from a high-level summary to increasingly granular details within the same visual, without needing separate pages or visuals.

How it improves exploration:

- Users start at an aggregate level (e.g., Country) and click to go deeper (Region → Category).
- One visual handles multiple levels of data granularity, reducing clutter.
- Business users can explore on their own without technical help.
- Filters and context are preserved as you drill deeper.

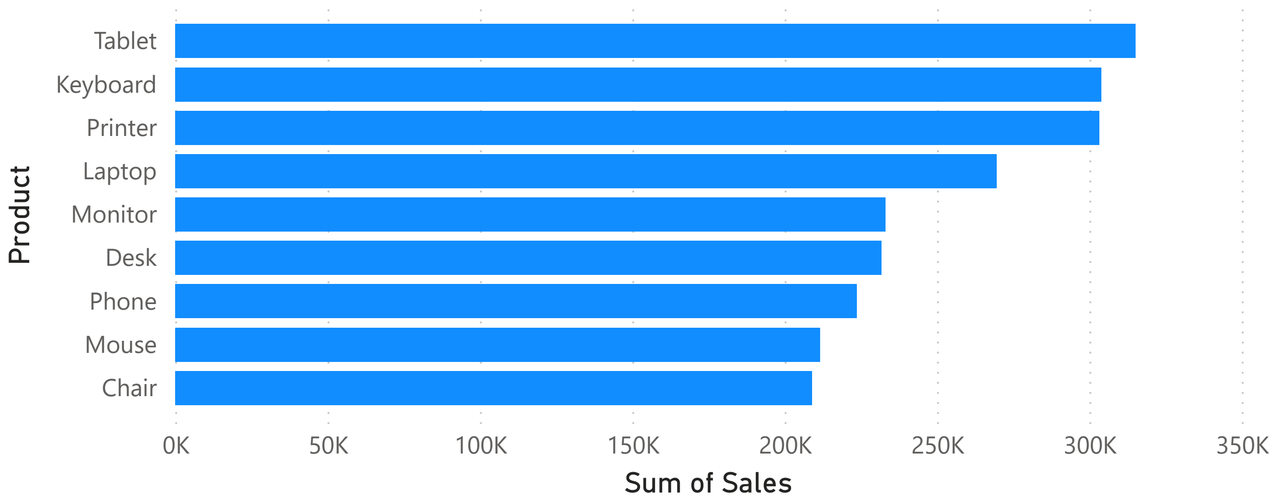
Example: A bar chart shows total sales by Category. Clicking "Electronics" drills into Laptop, Phone, Tablet, etc.

Q7. Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels.

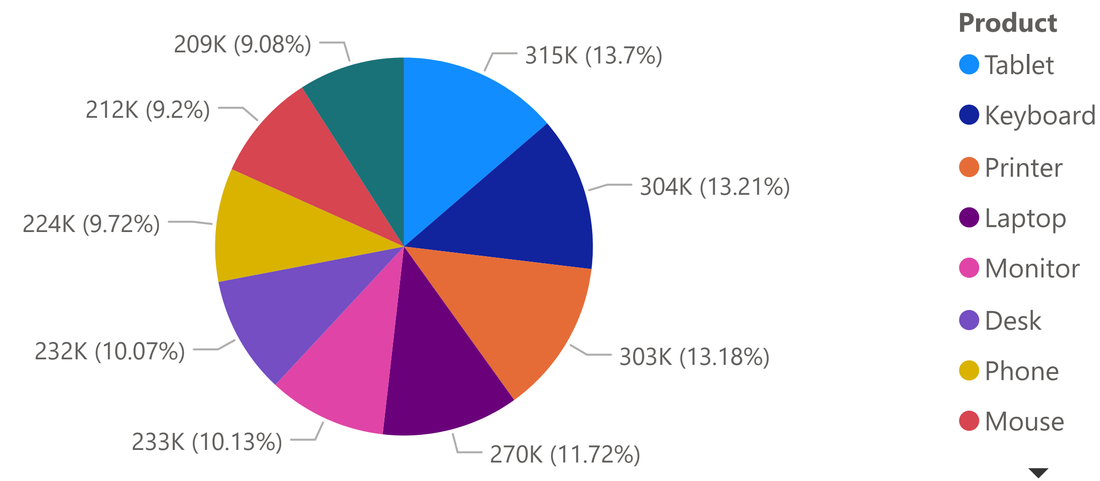


Q8. Implement Drill-Through so that selecting a Product navigates to a detailed report page showing regionwise sales.

Sum of Sales by Product

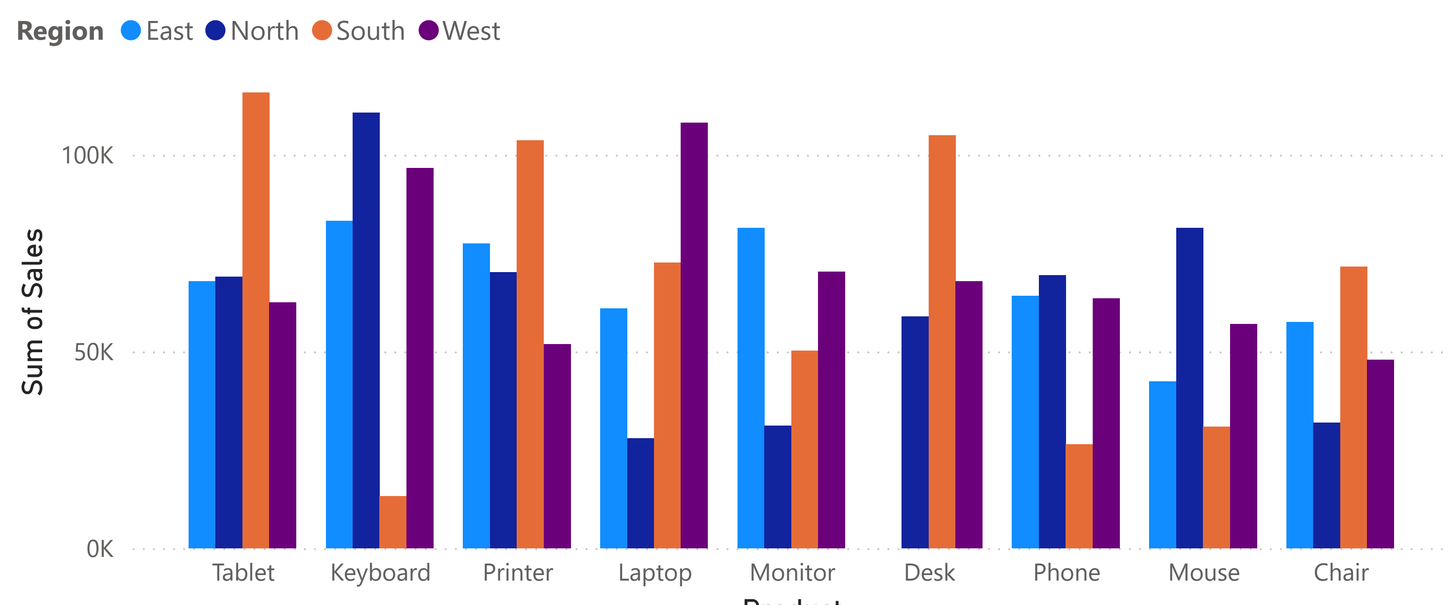


Sum of Sales by Product

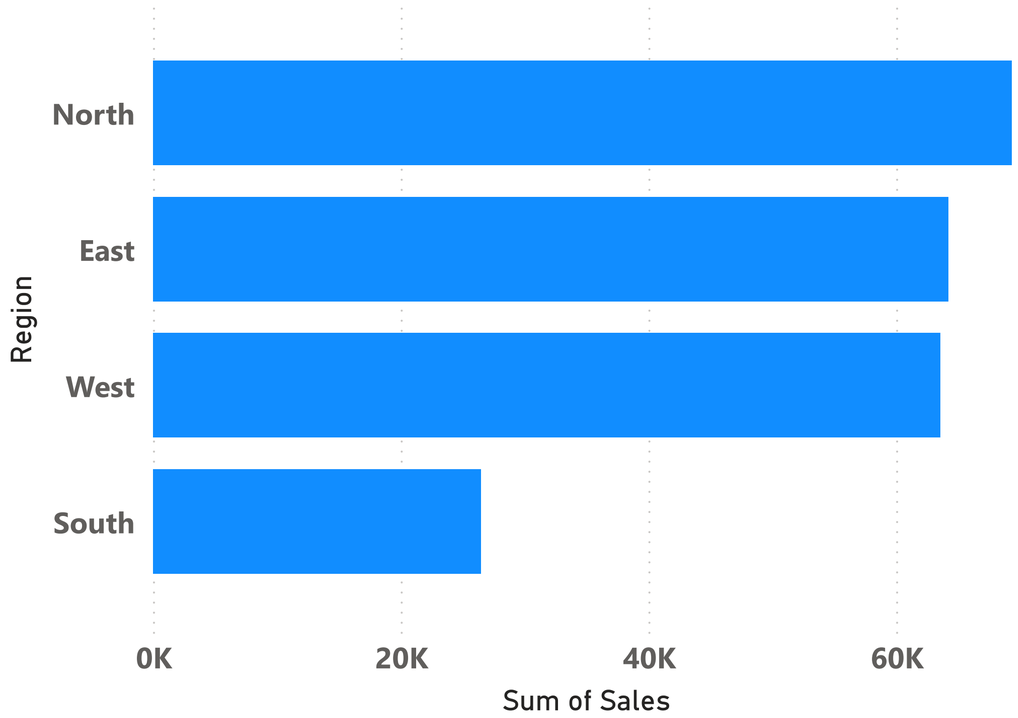


Product	Category	Sum of Quantity	Sum of Sales
Chair	Furniture	104	208996
Desk	Furniture	105	231793
Keyboard	Accessories	145	303877
Laptop	Electronics	160	269552
Monitor	Accessories	114	233033
Mouse	Accessories	116	211644
Phone	Electronics	103	223560
Printer	Office	151	303220
Tablet	Electronics	132	315168
Total		**1130**	**2300843**

Sum of Sales by Product and Region

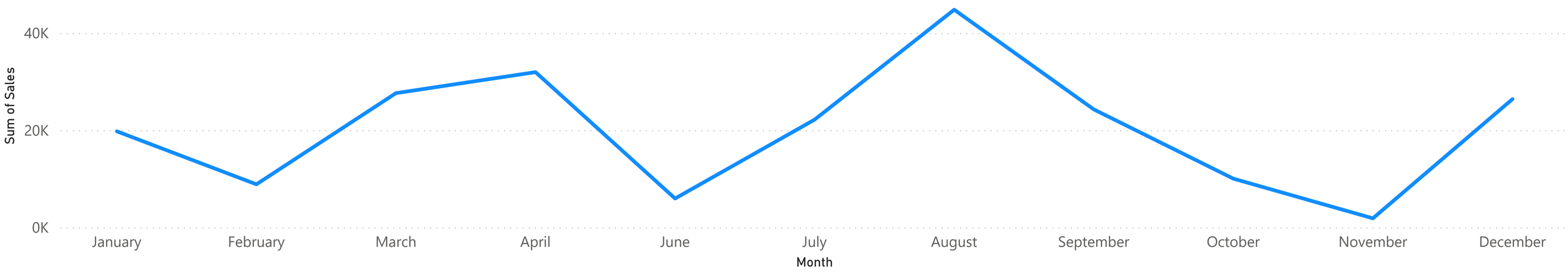


Sum of Sales by Region



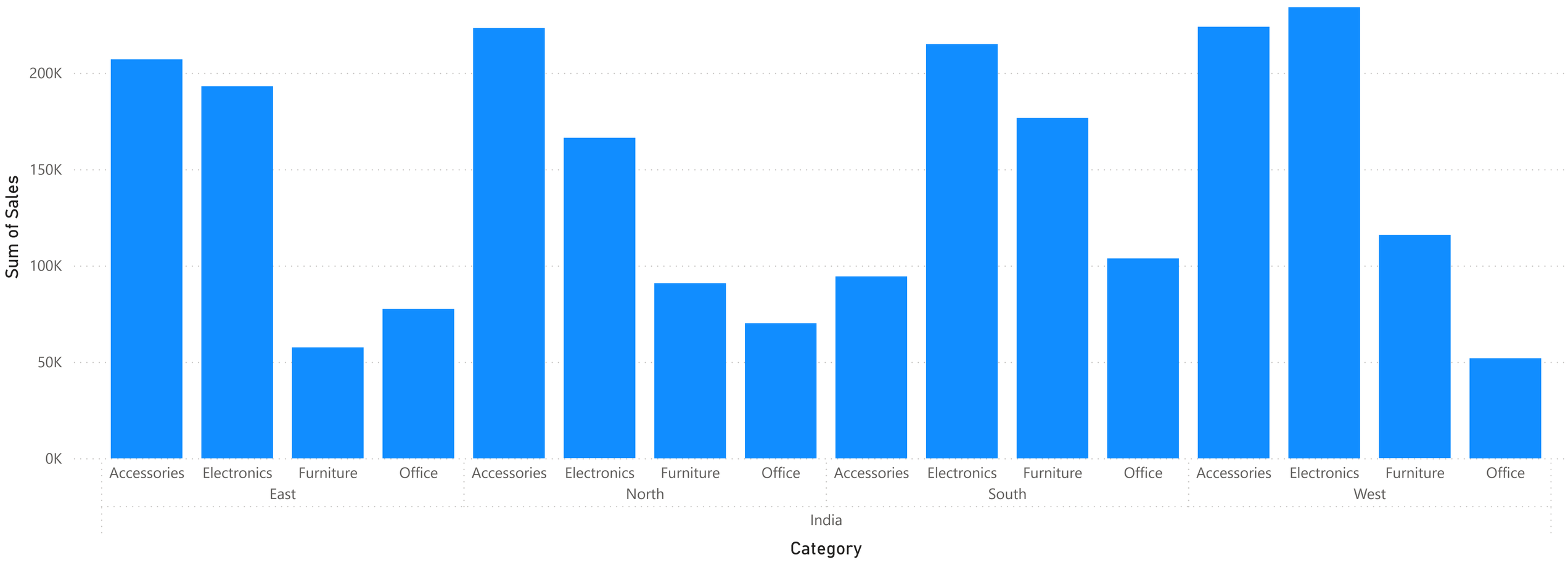
Order_ID	Year	Month	Customer_Name	Product	Region	Sum of Sales
1002	2023	February	Bhavya	Phone	South	800
1146	2022	January	Customer_126	Phone	South	19755
1160	2024	March	Customer_140	Phone	West	10709
1176	2024	March	Customer_156	Phone	East	7382
1181	2023	September	Customer_161	Phone	West	8443
1196	2022	November	Customer_176	Phone	North	1842
1203	2022	October	Customer_183	Phone	West	10010
1206	2024	March	Customer_186	Phone	West	4187
1216	2024	September	Customer_196	Phone	East	15828
1225	2023	April	Customer_205	Phone	West	4100
1228	2024	December	Customer_208	Phone	North	17850
Total						**223560**

Sum of Sales by Month



Q9. Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy.

Sum of Sales by Country, Region and Category



Q10. Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example.

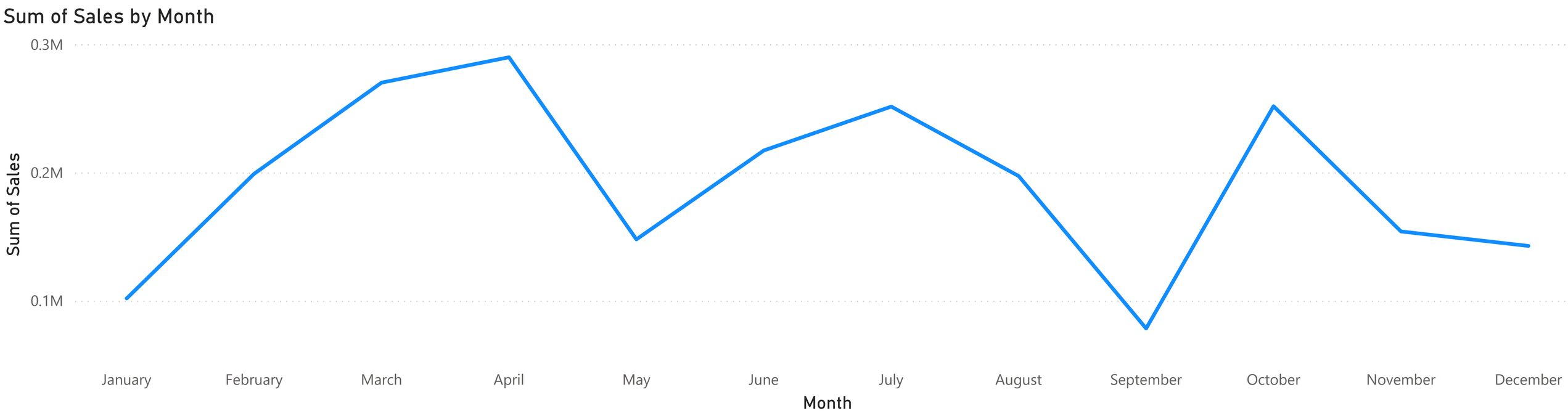
Answer 10

Drill-Down stays on the same page and same visual. You go deeper into the data. Like in our Sales dataset — I have a bar chart showing sales by Year. I click on 2023, it shows me Q1, Q2, Q3, Q4. I click Q3, it shows July, August, September. I never left the chart, just went deeper.

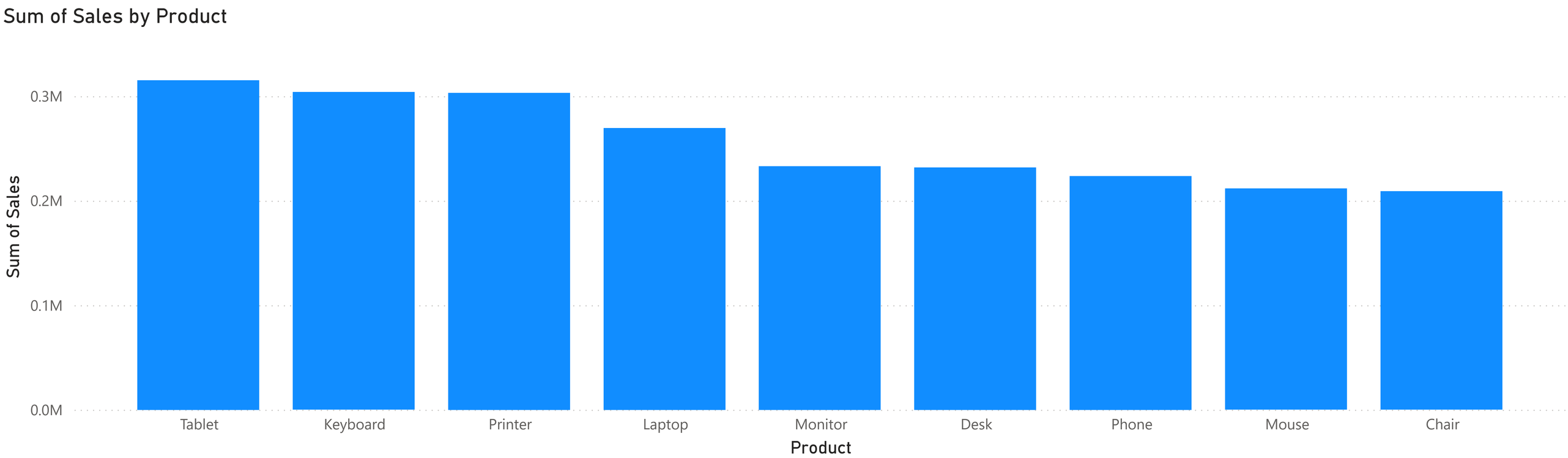
Drill-Through takes you to a completely different page. Like I right-click on "Laptop" in my product chart and it takes me to the Product Detail page showing only Laptop data. I moved from one page to another.

Simple real world example: Think of it like Google Maps. Drill-Down is like zooming in on the map — you're still on the same map, just seeing more detail. Drill-Through is like clicking on a restaurant and it opens a completely new page with reviews, menu, timings — different page, but related to what you clicked.

Q11. Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day.



Q12. Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000. Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column.



Q13 What is the “Keep All Filters” option in Drill-Through? How does it affect user interaction

Answer 13

When I drill-through from main page to Product Detail page, there are two scenarios:
Keep All Filters = OFF (default) Suppose on main page I have selected Region = North in slicer and then I drill through on Laptop. The Product Detail page will show Laptop data from ALL regions, not just North. Only the product filter carries over, everything else is ignored.

Keep All Filters = ON Same situation — I have Region = North selected and drill through on Laptop. Now the Product Detail page shows Laptop sales from North region only. All my slicer selections and filters travel with me to the new page.

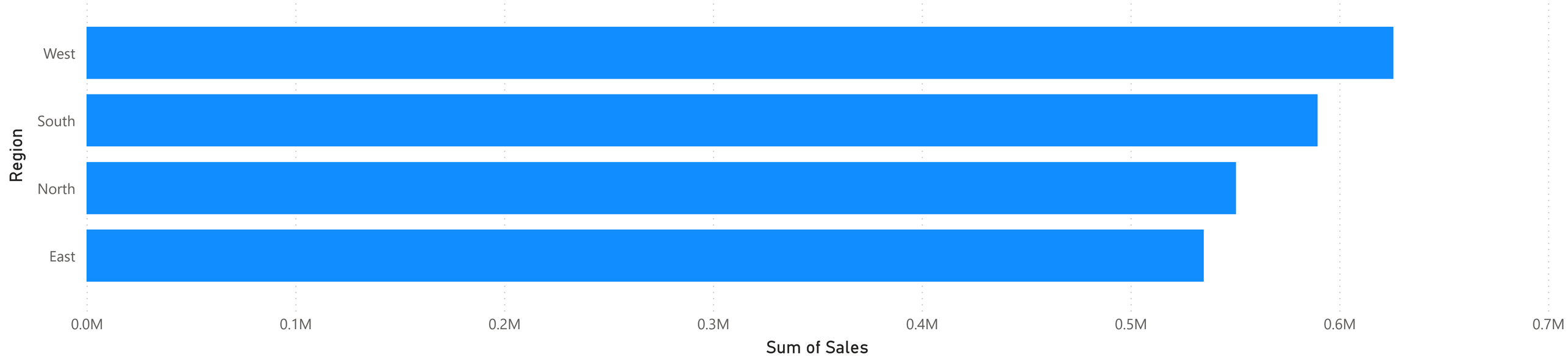
How to turn it on:

- Go to your drill-through destination page (Product Detail)
- In Visualizations pane, under Drill through section
- You'll see a toggle that says **"Keep all filters"**
- Turn it ON

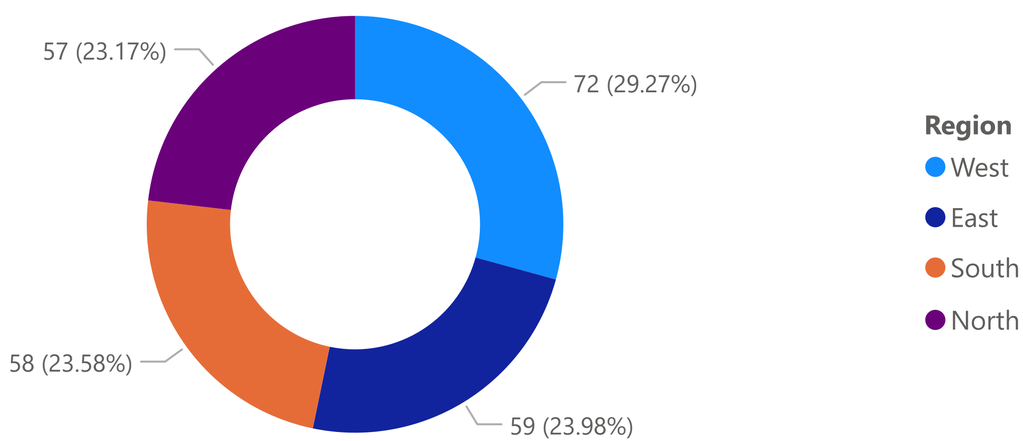
Which one to use: If you want users to see the complete picture of a product regardless of what they had selected, keep it OFF. If you want the context to be preserved, turn it ON. In our Sales dataset, turning it ON makes more sense because if someone was looking at North region and drills into Laptop, they probably want to see North region Laptop data specifically.

Q14 Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends over time.

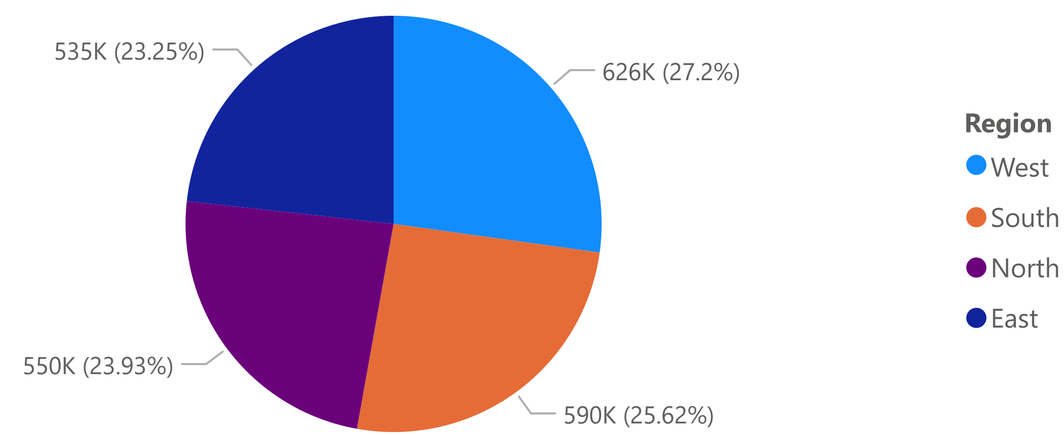
Sum of Sales by Region



Count of Order_ID by Region



Sum of Sales by Region

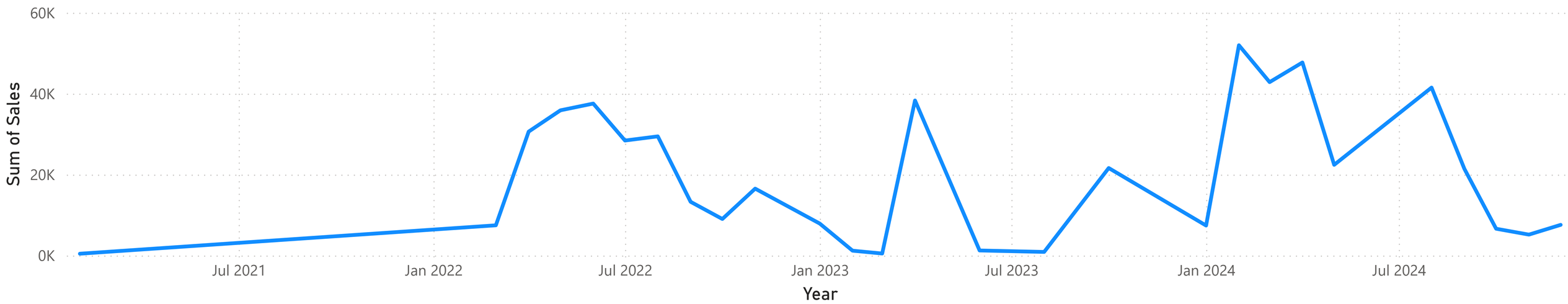


535K
Total Sales

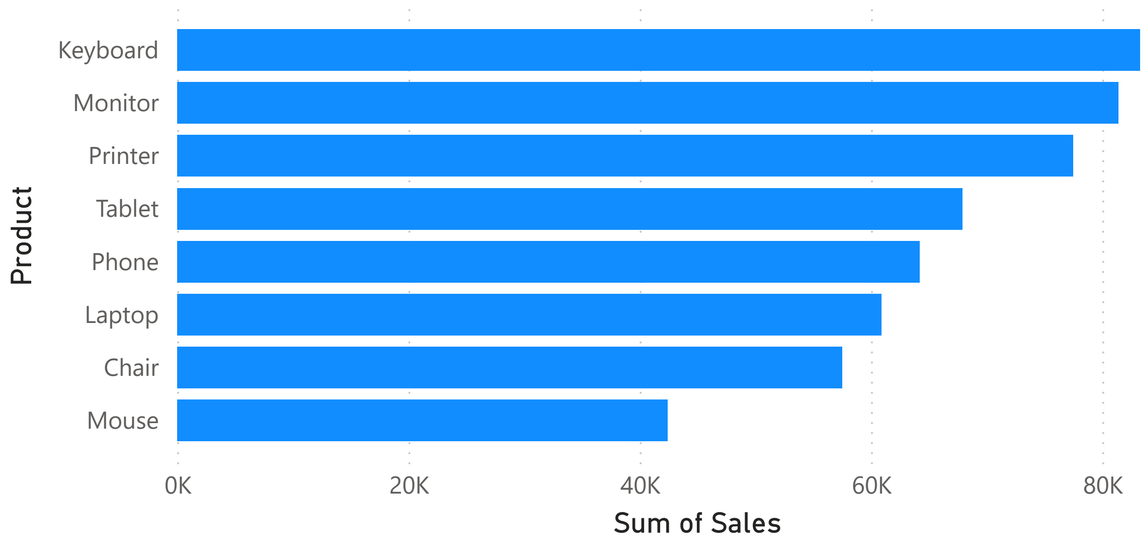
59
Count of Order_ID

9.07K
Average of Sales

Sum of Sales by Year, Quarter and Month



Sum of Sales by Product



Order_ID	Year	Quarter	Month	Day	Customer_Name	Product	Sum of Quantity	Sum of Sales
1003	2023	Qtr 1	March	20	Charan	Tablet	1	500
1007	2022	Qtr 3	July	15	Garima	Phone	2	850
1011	2023	Qtr 2	June	12	Kavya	Laptop	1	1250
1019	2021	Qtr 1	February	25	Tanmay	Monitor	3	450
1021	2024	Qtr 2	May	10	Customer_1	Printer	8	5890
1022	2022	Qtr 2	May	2	Customer_2	Keyboard	8	17350
1023	2024	Qtr 2	May	21	Customer_3	Keyboard	6	16523
1027	2024	Qtr 2	April	11	Customer_7	Printer	5	18412
1035	2022	Qtr 4	October	23	Customer_15	Chair	5	9029
1038	2023	Qtr 3	August	16	Customer_18	Tablet	3	891
1043	2024	Qtr 3	August	18	Customer_23	Tablet	3	15448
Total							**275**	**535012**

2M
Total Sales

246
Total Orders

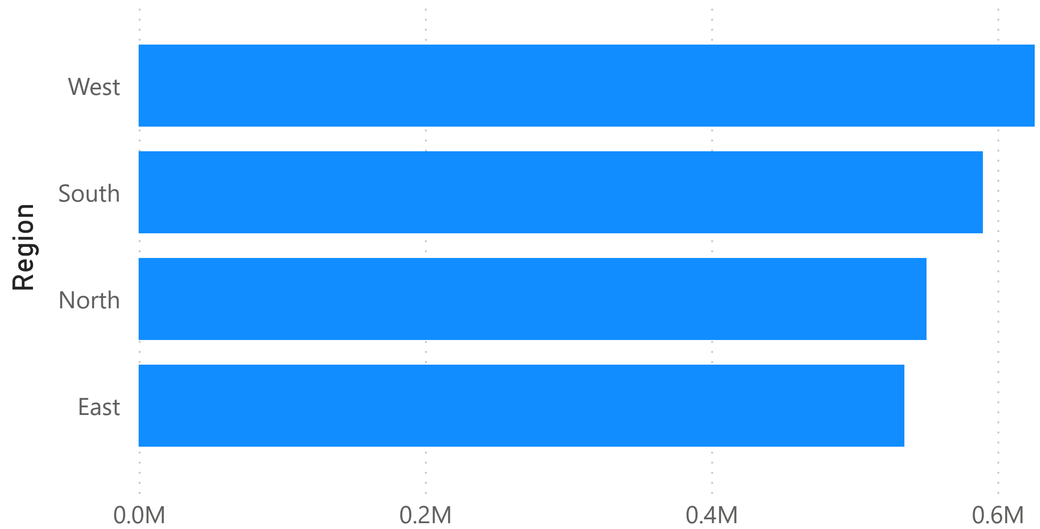
1K
Total Quantity

Year
- 2021
- 2022
- 2023
- 2024

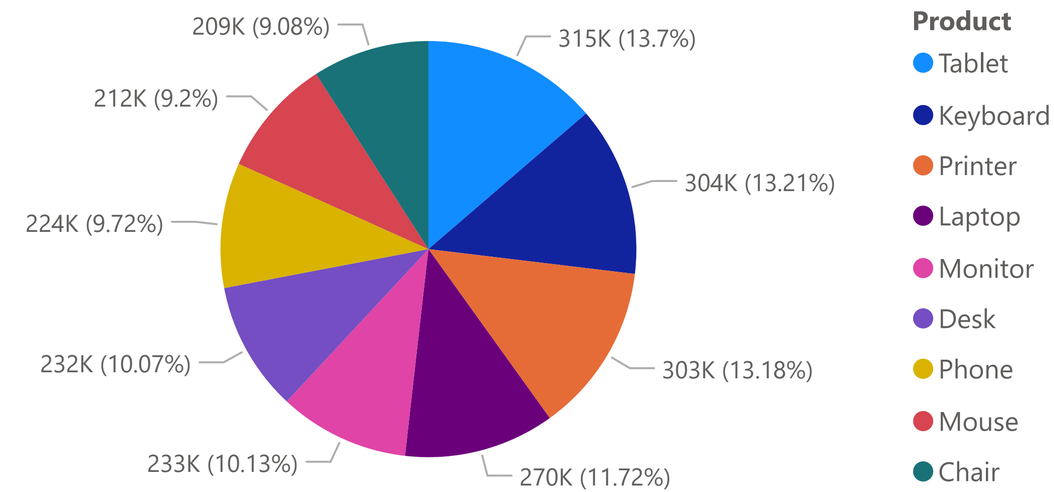
Category
- Accessories
- Electronics
- Furniture
- Office

Region
- East
- North
- South
- West

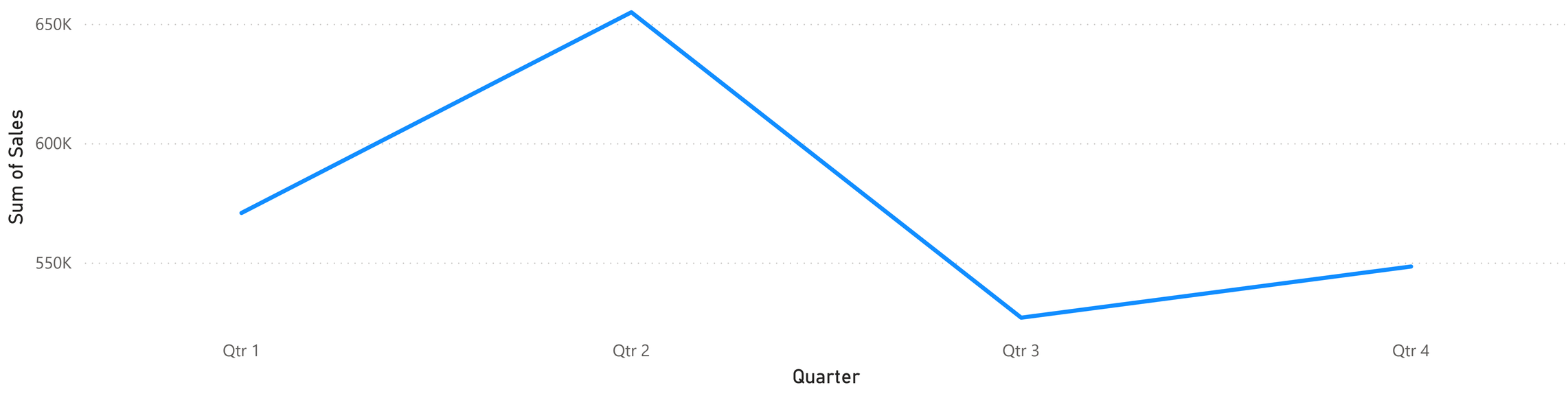
Sum of Sales by Region



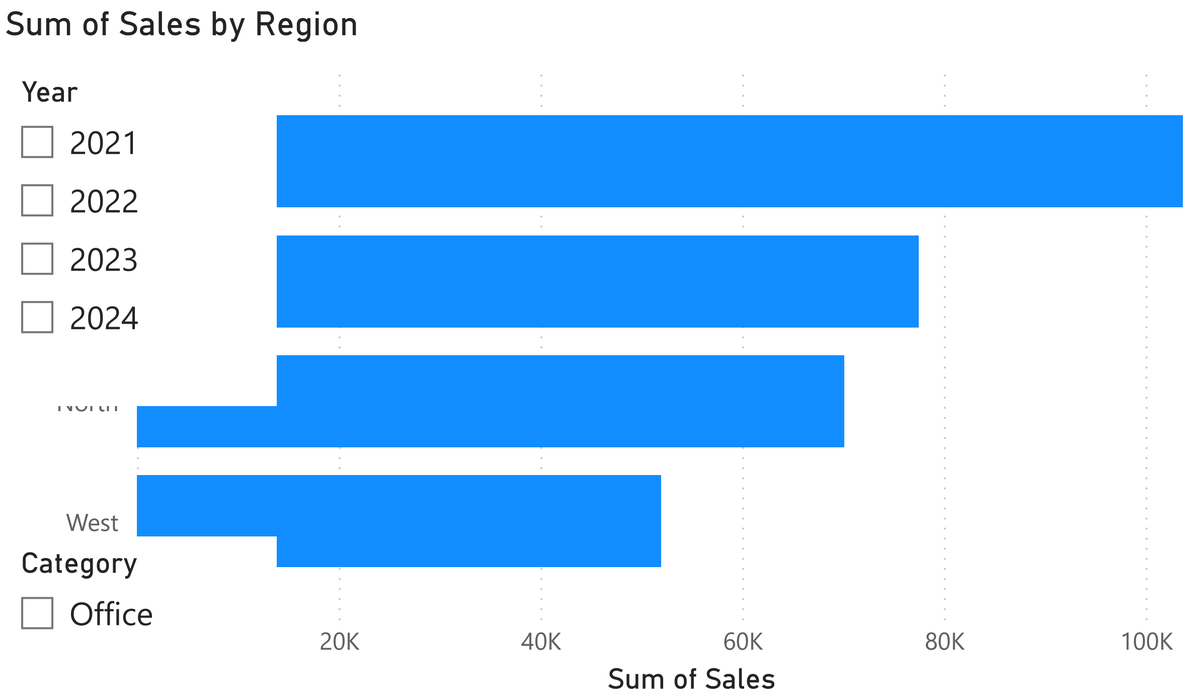
Sum of Sales by Product



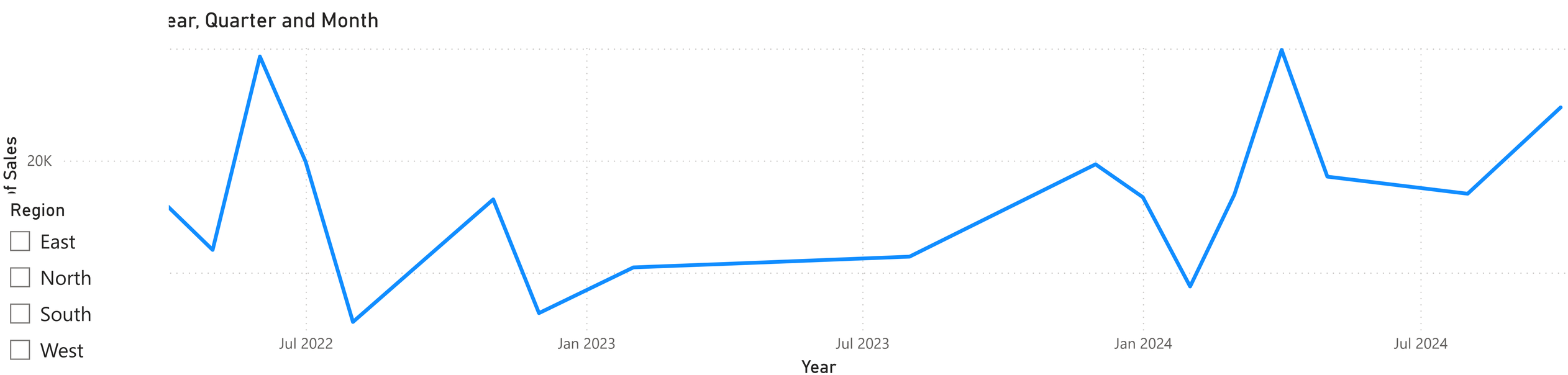
Sum of Sales by Quarter



Product Detail



Order_ID	Customer_Name	Year	Month	Region	Product	Sum of Quantity	Sum of Sales
1021	Customer_1	2024	May	East	Printer	8	5890
1027	Customer_7	2024	April	East	Printer	5	18412
1047	Customer_27	2022	August	East	Printer	4	2178
1048	Customer_28	2023	August	North	Printer	9	11421
1050	Customer_30	2022	July	South	Printer	9	1806
1065	Customer_45	2022	May	West	Printer	1	9708
1074	Customer_54	2023	February	North	Printer	3	5235
1083	Customer_63	2022	May	South	Printer	6	2316
1085	Customer_65	2024	August	South	Printer	2	17038
1090	Customer_70	2024	April	East	Printer	6	11466
1094	Customer_74	2024	February	West	Printer	4	1376
1097	Customer_77	2024	January	South	Printer	7	8808
1101	Customer_81	2022	June	East	Printer	5	2179
1104	Customer_84	2022	December	South	Printer	6	6381
Total						**151**	**303220**



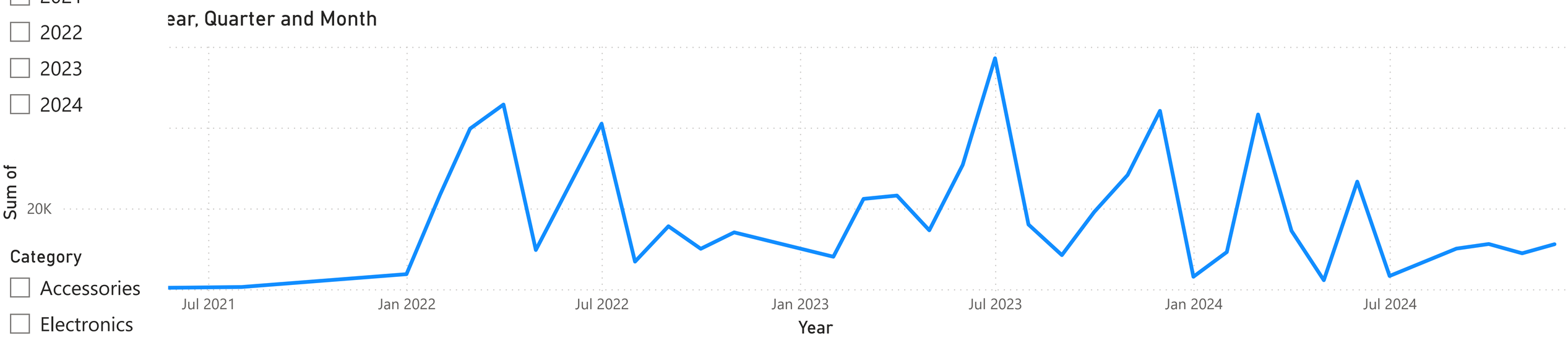
Region Detail

626K

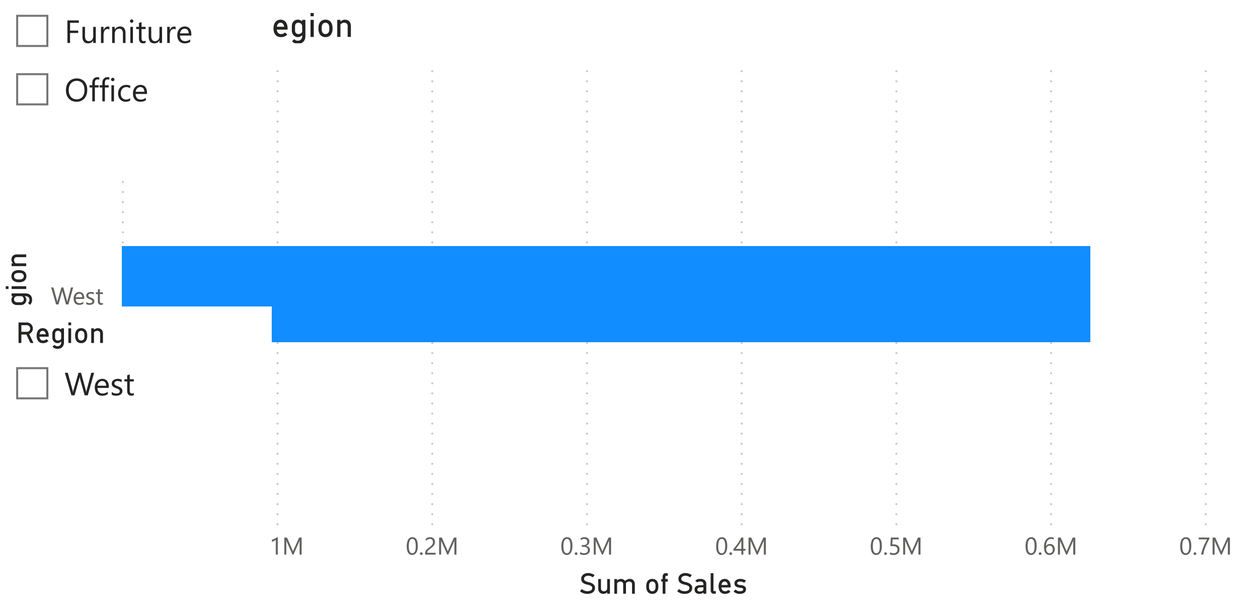
72
Count of Order_ID

8.69K
Average of Sales

Year
- 2021
- 2022
- 2023
- 2024



Category
- Accessories
- Electronics
- Furniture
- Office



Region
- West

Order_ID	Year	Month	Customer_Name	Region	Product	Sum of Quantity	Sum of Sales
1004	2023	April	Devansh	West	Monitor	3	300
1008	2021	August	Harsh	West	Tablet	1	550
1012	2023	July	Lakshay	West	Phone	2	900
1016	2022	November	Prisha	West	Laptop	1	1350
1020	2021	March	Ujjwal	West	Keyboard	5	125
1024	2024	February	Customer_4	West	Keyboard	5	2933
1025	2023	April	Customer_5	West	Laptop	9	6896
1031	2022	July	Customer_11	West	Tablet	2	3652
1033	2023	December	Customer_13	West	Monitor	2	7373
1034	2022	September	Customer_14	West	Monitor	7	12034
1036	2022	August	Customer_16	West	Desk	9	6831
1037	2022	June	Customer_17	West	Laptop	8	2527
Total						**315**	**625797**